Shangyong Shi

Pim Postdoc Fellow, Johns Hopkins University

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RESEARCH INTERESTS

- Snow hydrology, hydrometeorology, precipitation phase, snow-to-precipitation ratio, extreme precipitation
- Remote sensing, satellite precipitation retrieval
- Land surface modeling, machine learning, climate change

EDUCATION

Florida State University (FSU), Tallahassee, FL
Ph.D., Meteorology. Advisor: Guosheng Liu *Jan. 2021 – Aug. 2024*

Florida State University, Tallahassee, FL
M.S., Meteorology. Advisor: Guosheng Liu *Sept. 2018 – Dec. 2020*

Nanjing University (NJU), Nanjing, China
B.S., Atmospheric Sciences *Sept. 2014 – Jun. 2018*

National Taiwan University, Taipei, China
Exchange Student, Department of Atmospheric Sciences *Sept. 2016 – Jan. 2017*

FUNDINGS & AWARDS

- Glenadore and Howard L. Pim Postdoctoral Fellowship in Global Change ($60, 000/yr for 2 yrs), Department of Earth & Planetary Sciences (EPS) at Johns Hopkins University (JHU) 2024
- 1st place oral presentation among student entries in the Hydrology section, 2023 AMS 2023
- National Scholarship for outstanding undergraduates (top 2% in NJU) 2017
- The Liao's Scholarship (University-level, top 2% in school, NJU) 2016
- The Liao's Scholarship (University-level, top 2% in school, NJU) 2015
- University-level outstanding students (top 5% in NJU) 2015

EMPLOYMENTS AND EXPERIENCES

Johns Hopkins University, Department of Earth and Planetary Sciences Baltimore, MD
Pim Postdoc Fellow *Sept. 2024 - Present*
Mentor: Benjamin Zaitchik, Harihar Rajaram

- Implementing novel precipitation phase partitioning scheme in the Noah-MP model and test the sensitivity of snow accumulation to snow-to-precipitation (S/P) ratio.
- Exploring the impact of varying S/P ratio on streamflow.

University of Maryland,
Cooperative Institute for Satellite Earth System Studies College Park, MD
Research Intern *Jun 2023 – Aug 2023, Oct 2023 – May 2024*
Advisor: Yongzhen Fan, Huan Meng

- Developed an orographic precipitation index to identify orographic snowfall.
- Incorporate new variables in the machine learning algorithm to reduce the orographic snowfall rate bias estimates from satellite microwave sensors.

Florida State University,
Department of Earth, Ocean, and Atmospheric Science Tallahassee, FL
Research Assistant *Sept. 2018 – Aug. 2022, Jun. 2023 – Aug. 2024*

- Explore the role of extreme precipitation in Florida's hydroclimate variations.
- Investigated the trends in snow event to precipitation event ratio.
- Developed an energy-based phase partitioning scheme for satellite precipitation retrievals and hydrological modeling.
- Investigated the sensitivity of S/P ratio to warming using satellite data.

Florida State University,
Department of Earth, Ocean, and Atmospheric Science Tallahassee, FL
Teaching Assistant *Sept. 2022 – May 2023*

- Course: Atmospheric Dynamics I and II. Assisted syllabus design, guided recitation and conducted tank experiments.

Nanjing University, School of Atmospheric Sciences Nanjing, China
Research Assistant, Dissertation *Sept. 2017 – Jun. 2018*

- Studied the modification on the Indo-Western Pacific Ocean Capacitor Effect by the Pacific Meridional Mode in boreal spring.

Student Innovative Project Leader *Sept. 2015 – Jul. 2016*

- Simulated the Fujiwara Effect between two vortices in a rotating water tank.

PUBLICATIONS

1. **Shi, S.*,** & Liu, G. (2025). Investigation on the sensitivity of the snow-to-precipitation ratio to temperature based on satellite data (To be submitted to Water Resource Research)
2. **Shi, S.,** & Zaitchik, B. (2025). Snow modeling uncertainty induced by precipitation phase partitioning schemes (In preparation).
3. **Shi, S.,** Fan, Y., Dong, J., and Meng, H (2025). Developing a machine learning algorithm to improve orographic snowfall retrieval from satellite passive microwave sensors. (In preparation)
4. **Shi, S.***, & Liu, G. (2024). Improvements on Phase Classification Using Atmospheric Melting and Refreezing Energy Based on Soundings. *Journal of Geophysical Research: Atmospheres,* 129(10), e2023JD040030. https://doi.org/10.1029/2023JD040030.
5. Jeoung, H., **Shi, S.,** & Liu, G.* (2022). A novel approach to validate satellite snowfall retrievals by ground-based point measurements. *Remote Sensing*, 14(3), 434. https://doi.org/10.3390/rs14030434
6. **Shi, S.*,** & Liu, G. (2021). The latitudinal dependence in the trend of snow event to precipitation event ratio. *Scientific Reports*, 11(1), 18112. https://doi.org/10.1038/s41598-021-97451-9
7. **Shi, S.,** & Misra, V.*. (2020). The role of extreme rain events in Peninsular Florida's seasonal hydroclimate variations. *Journal of Hydrology*, 589, 125182. https://doi.org/10.1016/j.jhydrol.2020.125182

PRESENTATIONS

1. 2024 Dec. (Poster) Satellite Observed Non-linear Sensitivity of Snow-to-Precipitation Ratio to Temperature Warming. *2024 AGU Annual Meeting*.

2. 2024 Oct. (Invited talk) On the sensitivity of precipitation phase to temperature change. *JHU EPS A&O meeting.*
3. 2024 Jan. (Online talk) Developing a machine learning algorithm to improve orographic snowfall retrieval from satellite passive microwave sensors. *JPSS Hydrology Initiative Telecon*
4. 2023 Dec. (Poster) Improvements on Phase Classification Using Atmospheric Melting and Refreezing Energy Based on Soundings. *2023 AGU Annual Meeting*
5. 2023 Jan. (Oral) Classifying precipitation phase with atmospheric soundings. *2023 AMS Annual Meeting*

SERVICES

- Reviewer of Journal of Hydrology, Climate Dynamics, Asia-Pacific Journal of Atmospheric Sciences.
- Newsletter editor for Chinese-American Oceanic and Atmospheric Association

SKILLS

- Coding: Python, MATLAB, Fortran, C, GitHub code management
- Platforms: Linux, macOS, HPC,